Adaptive Stochastic Domination for Yang–Mills Bad Blocks

The exact quantifier missing from connected-event envelopes

20 July 2026

Abstract

We isolate and prove the probabilistic conversion theorem required by a dependent Yang–Mills bad-block programme. If an ordered Boolean bad field has next-site conditional probability bounded by q_i after every positive-probability exposure history, then a shared-uniform sequential coupling places it below independent Bernoulli variables of parameters q_i. This yields joint and connected-cluster bounds without constructing Mayer/Ursell activities. A two-site anticorrelated law proves that bounds on all intersection moments alone do not imply stochastic domination. At the Stage 14 effective weight, the abstract degree-18 cluster gate is subcritical. The remaining task is the full nonlinear SU(2) adaptive-history estimate.

1 Sequential coupling theorem

Let $X = (X_1, \dots, X_N) \in \{0,1\}^N$ have an arbitrary law. Fix an exposure order. For every history $h = (x_1, \dots, x_{i-1})$ of positive probability, write

$$p_i(h) = \mathbb{P}(X_i = 1 \mid X_1 = x_1, \dots, X_{i-1} = x_{i-1}).$$

Theorem 1.1 (Adaptive Bernoulli domination)**.** *If $p_i(h) \le q_i$ for every i and every positive-probability history, then there is a coupling with independent $Y_i \sim \mathrm{Bernoulli}(q_i)$ such that*

$$X_i \le Y_i \quad \textit{for every } i \quad \textit{almost surely.}$$

Consequently the law of X is stochastically dominated by the product law of Y.

Proof. Take independent $U_i \sim \mathrm{Uniform}[0,1]$. Generate X sequentially by

$$X_i = \mathbf{1}\{U_i \le p_i(X_1, \dots, X_{i-1})\}.$$

This reproduces the prescribed conditional probabilities and hence the law of X. On the same space put $Y_i = \mathbf{1}\{U_i \le q_i\}$. The Y_i are independent, and $p_i \le q_i$ gives $X_i \le Y_i$ pointwise. □

The premise is history-uniform, not merely an unconditional one-site or joint moment bound. It is also compatible with adaptive block selection: the next site may be a measurable function of the exposed history, provided the same conditional ceiling holds.

2 Connected clusters

Assume $q_i \le q$ and the bad-block adjacency graph has maximum degree Δ. The coupling gives, for every finite set S,

$$\mathbb{P}(X_i = 1 \ \forall i \in S) \le q^{|S|}.$$

Using the declared rooted connected-set count $\Delta^{2(n-1)}$, one obtains

Corollary 2.1. *If $e^{\alpha}\Delta^2 q < 1$, the weighted sum of bad connected sets containing a fixed root is at most*

$$\frac{e^{\alpha} q}{1 - e^{\alpha}\Delta^2 q}. \tag{1}$$

This is a stochastic-domination route. No partition-function activity is identified or required for the support-cluster statement.

3 Why intersection moments are insufficient

Proposition 3.1 (Two-site counterexample)**.** *Let*

$$\mathbb{P}(X = (1,0)) = \mathbb{P}(X = (0,1)) = \frac{1}{2}.$$

For $q = 1/2$, every nonempty intersection satisfies $\mathbb{P}(X_i = 1 \ \forall i \in S) \le q^{|S|}$. Nevertheless the law is not stochastically dominated by independent Bernoulli$(1/2)$ variables.

Proof. The singleton probabilities equal $1/2$ and the double intersection has probability zero. But the increasing event $\{X_1 = 1 \text{ or } X_2 = 1\}$ has probability one, whereas its product-law probability is $3/4$. Equivalently, $\mathbb{P}(X_2 = 1 \mid X_1 = 0) = 1$. □

Thus the connected-event envelopes from earlier stages do not alone justify stochastic domination. The adaptive conditional quantifier is essential.

4 Yang–Mills budget and frontier

At the Stage 14 sufficient parameters,

$$q_{\text{eff}} = e^{\kappa - J} + e^{-K} = 0.00243027750184,$$

while for $\alpha = 0.1$ and $\Delta = 18$,

$$q_{\text{crit}} = e^{-\alpha}\Delta^{-2} = 0.00279270808036.$$

The branching ratio is 0.870222534, and (1) is 0.0206959813. Therefore the event-to-cluster conversion closes exactly *if* every adaptive SU(2) block history obeys the stated conditional bound.

That premise remains open. The corrected frontier is to construct a gauge-covariant multi-link exposure retaining sufficient boundary information to charge center and continuous collar frustration, and to prove

$$\mathbb{P}(\text{next block bad} \mid \text{exposed history}) \le 0.00243027750184$$

uniformly in volume. Repeated RG, continuum construction, and the mass gap remain downstream. The present theorem is C2.